CALCULUS AND KINEMATICS ON ONE PAGE

Velocity to displacement, by parts

Repeat the product-rule identity, solve the recurring integral, and see where the half comes from.

Constant acceleration a; initial velocity u.

$$v(\tau) = u + a\tau, \qquad s(t) = \int_0^t v(\tau)\,\mathrm{d}\tau.$$

Here $s(t) = x(t) - x(0)$, and $t \geq 0$ is elapsed time.

The product rule, rearranged and integrated

$$\boxed{\int_0^t fg'\,\mathrm{d}\tau = \big[fg\big]_0^t - \int_0^t f'g\,\mathrm{d}\tau.}$$

Primes mean differentiation with respect to τ.

1 First pass: extract a boundary term

Take $f = v(\tau) = u + a\tau$ and $g' = 1$. Then $f' = a$ and $g = \tau$:

$$\begin{aligned} s(t) &= \big[\tau v(\tau)\big]_0^t - \int_0^t \tau a\,\mathrm{d}\tau \\ &= t(u + at) - aJ, \end{aligned}$$

where $J = \int_0^t \tau\,\mathrm{d}\tau$ is the remaining integral.

2 Repeat parts on the remaining integral

For J, choose $f = \tau$ and $g' = 1$. Now $f' = 1$ and $g = \tau$:

$$\begin{aligned} J &= \big[\tau \cdot \tau\big]_0^t - \int_0^t 1 \cdot \tau\,\mathrm{d}\tau \\ &= t^2 - J. \end{aligned}$$

The right-hand integral is exactly the same J. Collect the two copies:

$$2J = t^2 \qquad \Longrightarrow \qquad \boxed{J = \frac{t^2}{2}.}$$

3 Put the correction back

$$\begin{aligned} s(t) &= t(u + at) - \frac{1}{2}at^2 \\ &= \boxed{ut + \frac{1}{2}at^2.} \end{aligned}$$

Equivalently, $s = \dfrac{u + v(t)}{2}\,t$: **mean velocity** $\times$ **elapsed time**.

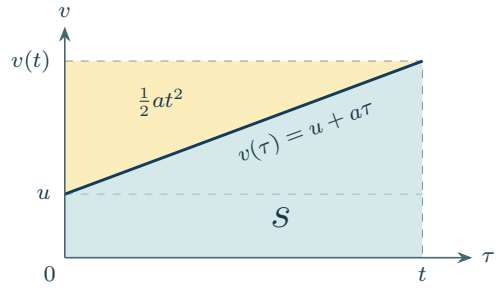
The repeated-parts table

Differentiate v on the left; repeatedly integrate 1 on the right.

sign	differentiate v	integrate 1
$+$	$u + a\tau$	τ
$-$	a	$\tau^2/2$
stop	0	$\tau^3/6$

Add the signed row products; evaluate between 0 and t.

A rectangle, then a correction



Rectangle $t\,v(t)$ minus the gold triangle:

$$s = t\,v(t) - \frac{1}{2}at^2.$$

Sketch: $u, a, t > 0$. In general, s is signed.

The lower endpoint

At $\tau = 0$, both $\tau v(\tau)$ and τ^2 vanish.

A numerical check

$u = 4\,\mathrm{m\,s^{-1}}$, $a = 2\,\mathrm{m\,s^{-2}}$, $t = 3\,\mathrm{s}$.

Then $v(t) = 4 + 2(3) = 10\,\mathrm{m\,s^{-1}}$.

$$s = \underbrace{30\,\mathrm{m}}_{t\,v(t)} - \underbrace{9\,\mathrm{m}}_{at^2/2} = \boxed{21\,\mathrm{m}}.$$

One more term means one more row

Let $v(\tau) = u + a_0\tau + b\tau^2$, with u, a_0, b constant. Its derivatives are v, $a_0 + 2b\tau$, $2b$, 0:

$$s = t(u + a_0 t + bt^2) - \frac{t^2}{2}(a_0 + 2bt) + \frac{t^3}{6}(2b) = \boxed{ut + \frac{1}{2}a_0 t^2 + \frac{1}{3}bt^3.}$$

Each by-parts step differentiates one factor and integrates the other. For polynomial velocity, a zero derivative makes the process finite.